

MINERAL MONITOR

Africa-Focused Mineral Intelligence Platform

Project Scope & Technical Documentation | Version 1.0

1. Executive Summary

Mineral Monitor (MM) is an intelligence platform purpose-built for mineral investment across the African continent. It aggregates fragmented data satellite imagery, GIS deposit records, geopolitical risk indexes, concession registries, macroeconomic indicators, and real-time news into a single, AI-powered decision-support environment.

The platform targets institutional investors, multilateral development banks, commodity traders, and policy institutions such as the World Bank, African Development Bank (ADB), ECOWAS, and private equity firms active in the extractive sector. MM delivers what no current single platform provides: a consolidated, Africa-centric mineral intelligence layer that is analytically rigorous, geospatially precise, and continuously updated.

Platform Positioning

Mineral Monitor sits at the convergence of financial intelligence, geospatial analytics (GIS/MCAS platforms), news, and AI-driven research synthesis serving as the primary discovery and due-diligence layer for mineral investment.

2. Problem Statement

Africa holds an estimated 30% of the world's mineral reserves, yet investment decisions are routinely hampered by fragmented, inaccessible, and inconsistent data infrastructure. Key challenges include:

- Mineral deposit data is scattered across national GIS portals, MCAS databases, and bilateral agency reports with no unified access point.
- Geopolitical and regulatory risk signals exist in isolation from deposit-level data, forcing investors to manually correlate unrelated datasets.
- Concession status, operational history, and production returns are largely opaque to external investors, particularly for junior and greenfield assets.
- Satellite imagery and remote sensing outputs are rarely cross-referenced with news sentiment or macroeconomic indicators in real time.
- AI-driven synthesis tools in the commodity intelligence space remain predominantly focused on non-African markets.

MM directly addresses these gaps by functioning as an integrated intelligence layer — consolidating signals, surfacing opportunities, and enabling faster, more confident investment decisions.

3. Platform Overview & Core Objectives

MM is architected around five core intelligence objectives:

- **Worldview Insights:** Deliver aggregated, continent-scale visibility into mineral investment variables across all 54 African nations, with ECOWAS and West Africa as the primary regional focus.
- **Signal Consolidation:** Combine news, satellite, GIS, macroeconomic, and concession data into unified investment-grade outputs eliminating the need for manual cross-platform research.
- **AI-Driven Synthesis:** Generate automated intelligence reports that compound multiple data streams into coherent investment overviews, actionable alerts, and trend forecasts.
- **Concession Intelligence:** Maintain live status tracking of mineral concessions identifying underperforming, inactive, or opportunity-grade concessions requiring capital injection.
- **Institutional Accessibility:** Provide data interfaces consumable by Bloomberg terminals, World Bank data portals, ADB dashboards, and third-party research platforms via API.

4. Data Sources & Integration Architecture

The platform ingests and normalizes data across six primary source categories:

Data Module	Source / Integration	Refresh Cadence
International Bank Reports	World Bank, AfDB, IMF minerals & extractives briefs	Quarterly / On-publish
Economic Stability Indicators	GDP, inflation, FX reserves, trade balance (Africa-focused)	Monthly
Geopolitical Risk Indexes	Conflict monitors, political stability scores, ACLED, ICRG	Weekly / Real-time
News Intelligence	Wire services, local African press, regulatory announcements	Real-time
Satellite Imagery (vs. MCAS)	Multispectral & SAR imagery cross-referenced with MCAS baselines	15-day / On-demand
GIS Mineral Deposit Data	National cadastral data, historical deposit surveys (10yr+)	Continuous

5. Mineral Investment Variables

5.1 Geopolitical & Risk Intelligence

- Country-level and sub-national risk scores covering political stability, conflict exposure, regulatory volatility, and expropriation risk.
- Sentiment analysis applied to news streams: positive, negative, and neutral scoring with mineral-asset and jurisdiction tagging.
- Conflict event mapping correlated with proximity to active concessions and deposit clusters.

5.2 Deposit & Concession Intelligence

- Mineral deposit location database spanning 10 years to present, with depth, grade estimates, mineral type, and ownership history.
- Concession registry tracking: active, suspended, expired, available across jurisdictions.
- Production-based analysis: actual vs. projected output, investment returns, and operational efficiency benchmarks for producing assets.
- Opportunity identification: AI flagging of concessions with favourable geology, weak operational status, or available licensing grounded in derisking and status-of-operations analysis.

5.3 Satellite & Remote Sensing Analytics

- Historical and current satellite imagery (multispectral and SAR) overlaid against MCAS mineral occurrence baselines.
- Predictive trend modelling: satellite-derived signals combined with news sentiment to forecast mineral activity levels.

5.4 Macroeconomic Context

- Economic stability indicators (GDP growth, currency risk, debt service ratios) at national and regional levels.
- ECOWAS regional aggregates and cross-border investment flow data.
- International bank mineral sector reports and commodity price correlation analysis.

6. AI & Analytics Engine

The MM analytics engine sits at the core of the platform, performing four primary functions:

1. Report Consolidation

Ingests and synthesises reports from international banks, NGOs, and government agencies. Produces structured investment summaries by mineral type, country, and risk tier removing the burden of manual document review from analysts.

2. Sentiment Analysis Engine

Applies NLP models trained on African commodities and political contexts to score news articles, regulatory filings, and social signals. Outputs are mapped to specific concessions and deposit clusters, enabling near real-time risk pulse monitoring.

3. Predictive Mineral Trend Modelling

Combines satellite change detection outputs with sentiment scores, production data, and macroeconomic indicators to generate forward-looking trend projections at the deposit and regional level.

4. Investment Opportunity Scoring

Ranks concessions and deposit areas by a composite opportunity score factoring in geological prospectivity, operational status, risk index, concession availability, and required capital providing a prioritised pipeline for investor outreach.

7. Target Audience & Use Cases

MM is designed for institutional and professional users operating across the mineral investment value chain:

Data Module	Source / Integration	Refresh Cadence
Institutional Investors	Private equity, sovereign wealth funds, commodity desks	Investment screening & due diligence
Multilateral Banks	World Bank, AfDB, ECOWAS development arms	Sector monitoring & policy briefs
Bloomberg / Data Terminals	Financial data platforms & aggregators	API data feeds & research overlays
Mining Companies	Junior to major mining operators	Competitive intelligence & expansion analysis
Governments & Regulators	Mining ministries, cadastral authorities	Concession management & FDI attraction
Research Institutions	Think tanks, consultancies	African extractives research

8. Delivery Phases & Roadmap

Phase	Scope	Timeline
Phase 1	Core data ingestion pipeline: GIS deposit data, concession registry, news feeds, geopolitical indexes. Foundational dashboard with deposit map and risk overlay.	Months 1 – 4
Phase 2	Satellite imagery integration with MCAS cross-referencing. Sentiment analysis engine deployment. AI report consolidation module launch.	Months 5 – 8
Phase 3	Predictive trend modelling. Investment opportunity scoring engine. Production analytics and concession opportunity flags.	Months 9 – 12
Phase 4	API layer for Bloomberg / World Bank / AfDB integration. Enterprise dashboards. White-label and institutional data licensing.	Months 13 – 16

9. Scope & Boundaries

In Scope

- African continent mineral intelligence (primary focus: West Africa / ECOWAS).

- Investment variable data covering geopolitics, macroeconomics, deposits, concessions, satellite, and news.
- AI synthesis, sentiment analysis, and opportunity scoring.
- API and data feeds for third-party platform integration.

Out of Scope (Phase 1)

- Direct transaction facilitation or brokerage of mineral rights.
- Real-time commodity spot pricing (to be considered for Phase 3 API feeds).
- Non-African mineral jurisdictions in initial launch scope.

10. Conclusion

Mineral Monitor represents a transformative leap in how African mineral intelligence is accessed, synthesised, and acted upon. By consolidating historically fragmented data sources into a single, AI-powered platform, MM reduces investment risk, accelerates due diligence, and opens Africa's mineral economy to a broader, better-informed global investment community.

The platform is positioned as critical infrastructure for the next generation of African mineral investment serving institutions from Bloomberg terminals to development banks, and from sovereign investors to junior mining companies seeking their next producing asset.